

ASSIGNMENT 2 — OPERATIONAL SEMANTICS

COMP 3010 — ORGANIZATION OF PROGRAMMING LANGUAGES

Remember: For every answer (including multiple choice) explain *how* you got your answer and *why* you think it is the correct one. This explanation will be part of your grade.

1. BIG-STEP OPERATIONAL SEMANTICS

For exercises 1 to 3, consider this definition of conditional arithmetic in terms of a big-step operational semantics.

Syntax of arithmetic (A) and boolean (B) expressions, and natural number (n) and boolean (b) values:

$n ::= 0 \mid 1 \mid 2 \mid 3 \mid \dots$
 $A ::= \underline{n} \mid \text{plus}(A_1, A_2) \mid \text{minus}(A_1, A_2) \mid \text{times}(A_1, A_2) \mid \text{div}(A_1, A_2) \mid \text{if}(B, A_1, A_2)$
 $b ::= \text{true} \mid \text{false}$
 $B ::= \underline{b} \mid \text{and}(B_1, B_2) \mid \text{or}(B_1, B_2) \mid \text{zero?}(A)$

Big-step operational semantics of arithmetic expressions ($A \Downarrow n$):

$$\begin{array}{c}
 \overline{\underline{n} \Downarrow n} \\
 \frac{A_1 \Downarrow n_1 \quad A_2 \Downarrow n_2 \quad n_1 + n_2 = n}{\text{plus}(A_1, A_2) \Downarrow n} \quad \frac{A_1 \Downarrow n_1 \quad A_2 \Downarrow n_2 \quad n_1 - n_2 = n \quad n_1 \geq n_2}{\text{minus}(A_1, A_2) \Downarrow n} \\
 \frac{A_1 \Downarrow n_1 \quad A_2 \Downarrow n_2 \quad n_1 \times n_2 = n}{\text{times}(A_1, A_2) \Downarrow n} \quad \frac{A_1 \Downarrow n_1 \quad A_2 \Downarrow n_2 \quad n_1 \div n_2 = n \quad n_2 \neq 0}{\text{div}(A_1, A_2) \Downarrow n} \\
 \frac{B \Downarrow \text{true} \quad A_1 \Downarrow n_1}{\text{if}(B, A_1, A_2) \Downarrow n_1} \quad \frac{B \Downarrow \text{false} \quad A_2 \Downarrow n_2}{\text{if}(B, A_1, A_2) \Downarrow n_2}
 \end{array}$$

Big-step operational semantics of boolean expressions ($B \Downarrow b$):

$$\begin{array}{c}
 \overline{\underline{\text{true}} \Downarrow \text{true}} \quad \overline{\underline{\text{false}} \Downarrow \text{false}} \\
 \frac{B_1 \Downarrow \text{true} \quad B_2 \Downarrow b}{\text{and}(B_1, B_2) \Downarrow b} \quad \frac{B_1 \Downarrow \text{false}}{\text{and}(B_1, B_2) \Downarrow \text{false}} \\
 \frac{B_1 \Downarrow \text{false} \quad B_2 \Downarrow b}{\text{or}(B_1, B_2) \Downarrow b} \quad \frac{B_1 \Downarrow \text{true}}{\text{or}(B_1, B_2) \Downarrow \text{true}} \\
 \frac{A \Downarrow 0}{\text{zero?}(A) \Downarrow \text{true}} \quad \frac{A \Downarrow n \quad n \neq 0}{\text{zero?}(A) \Downarrow \text{false}}
 \end{array}$$

For the natural number division $n_1 \div n_2$ returns only the whole number dividend and drops the remainder, so that $7 \div 2$ is 3 for example.

Exercise 1 (Multiple Choice). Which of the following evaluations of

$$\text{times} \left(\text{if} \left(\text{or} \left(\text{zero?}(\underline{0}), \text{zero?}(\text{div}(\underline{0}, \underline{1})) \right), \underline{4}, \underline{2} \right), \text{minus}(\underline{5}, \underline{3}) \right)$$

can be derived by the operational semantics?

- (a) $\text{times} \left(\text{if} \left(\text{or} \left(\text{zero?}(\underline{0}), \text{zero?}(\text{div}(\underline{0}, \underline{1})) \right), \underline{4}, \underline{2} \right), \text{minus}(\underline{5}, \underline{3}) \right) \Downarrow 0$
- (b) $\text{times} \left(\text{if} \left(\text{or} \left(\text{zero?}(\underline{0}), \text{zero?}(\text{div}(\underline{0}, \underline{1})) \right), \underline{4}, \underline{2} \right), \text{minus}(\underline{5}, \underline{3}) \right) \Downarrow 2$
- (c) $\text{times} \left(\text{if} \left(\text{or} \left(\text{zero?}(\underline{0}), \text{zero?}(\text{div}(\underline{0}, \underline{1})) \right), \underline{4}, \underline{2} \right), \text{minus}(\underline{5}, \underline{3}) \right) \Downarrow 4$
- (d) $\text{times} \left(\text{if} \left(\text{or} \left(\text{zero?}(\underline{0}), \text{zero?}(\text{div}(\underline{0}, \underline{1})) \right), \underline{4}, \underline{2} \right), \text{minus}(\underline{5}, \underline{3}) \right) \Downarrow 8$
- (e) $\text{times} \left(\text{if} \left(\text{or} \left(\text{zero?}(\underline{0}), \text{zero?}(\text{div}(\underline{0}, \underline{1})) \right), \underline{4}, \underline{2} \right), \text{minus}(\underline{5}, \underline{3}) \right) \Downarrow \text{true}$
- (f) $\text{times} \left(\text{if} \left(\text{or} \left(\text{zero?}(\underline{0}), \text{zero?}(\text{div}(\underline{0}, \underline{1})) \right), \underline{4}, \underline{2} \right), \text{minus}(\underline{5}, \underline{3}) \right) \Downarrow \text{false}$

Exercise 2 (This or That). An arithmetic expression A *returns* if there is some number n such that $A \Downarrow n$, and *diverges* if there is no such n . Similarly, a boolean expression B *returns* if there is some boolean value b such that $B \Downarrow b$, and *diverges* otherwise.

For each of the following arithmetic and boolean expressions, say if that expression returns or diverges and reason your way through using a derivation tree for each.

- (a) $\text{div}(\text{plus}(\underline{3}, \underline{1}), \text{minus}(\underline{5}, \underline{5}))$
- (b) $\text{if}(\text{and}(\underline{\text{false}}, \text{zero?}(\text{div}(\underline{1}, \underline{0}))), \text{div}(\underline{3}, \underline{0}), \underline{7})$
- (c) $\text{and}(\text{zero?}(\text{minus}(\underline{2}, \underline{3})), \underline{\text{true}})$
- (d) $\text{or}(\underline{\text{true}}, \text{zero?}(\text{div}(\underline{0}, \underline{0})))$
- (e) $\text{if}(\text{zero?}(\underline{0}), \text{div}(\underline{10}, \underline{2}), \text{plus}(\underline{1}, \text{div}(\underline{0}, \underline{0})))$

Exercise 3 (Show Your Work). Determine the number the following expression evaluates to by drawing a derivation tree of the big-step operational semantics:

$$\text{if} \left(\text{and} \left(\text{zero?}(\text{minus}(\text{plus}(\underline{2}, \underline{2}), \underline{4})), \underline{\text{true}} \right), \text{div}(\text{times}(\underline{6}, \underline{3}), \underline{3}), \text{div}(\underline{5}, \text{minus}(\underline{2}, \underline{2})) \right)$$

2. SMALL-STEP OPERATIONAL SEMANTICS

For exercises 4 to 6, consider this definition of a small-step operational semantics for the same conditional arithmetic language used previously in section 1.

Small-step reduction rules:

$$\begin{array}{llll} \text{plus}(n_1, n_2) \mapsto \underline{n} & (n = n_1 + n_2) & \text{minus}(n_1, n_2) \mapsto \underline{n} & (n = n_1 - n_2, n_1 \geq n_2) \\ \text{times}(n_1, n_2) \mapsto \underline{n} & (n = n_1 \times n_2) & \text{div}(n_1, n_2) \mapsto \underline{n} & (n = n_1 \div n_2, n_2 \neq 0) \\ \text{if}(\underline{\text{true}}, A_1, A_2) \mapsto A_1 & & \text{if}(\underline{\text{false}}, A_1, A_2) \mapsto A_2 & \\ \text{and}(\underline{\text{true}}, B) \mapsto B & & \text{and}(\underline{\text{false}}, B) \mapsto \underline{\text{false}} & \\ \text{or}(\underline{\text{false}}, B) \mapsto B & & \text{or}(\underline{\text{true}}, B) \mapsto \underline{\text{true}} & \\ \text{zero?}(\underline{0}) \mapsto \underline{\text{true}} & & \text{zero?}(\underline{n}) \mapsto \underline{\text{false}} & (n \neq 0) \end{array}$$

Evaluation contexts (E):

$$\begin{aligned}
 E ::= & \square \mid \text{plus}(E, A) \mid \text{plus}(\underline{n}, E) \mid \text{minus}(E, A) \mid \text{minus}(\underline{n}, E) \\
 & \mid \text{times}(E, A) \mid \text{times}(\underline{n}, E) \mid \text{div}(E, A) \mid \text{div}(\underline{n}, E) \mid \text{if}(E, A_1, A_2) \\
 & \mid \text{and}(E, B) \mid \text{or}(E, B) \mid \text{zero?}(E)
 \end{aligned}$$

Reducing sub-expressions is *only* allowed within evaluation contexts:

$$\frac{A \mapsto A'}{E[A] \mapsto E[A']} \qquad \frac{A \mapsto B'}{E[B] \mapsto E[B']}$$

Exercise 4 (Multiple Choice). What do you get from plugging the expression $\text{minus}(\text{plus}(\underline{2}, \underline{1}), \underline{3})$ into the evaluation context $\text{if}(\text{zero?}(\square), \text{times}(\underline{3}, \underline{5}), \text{plus}(\underline{1}, \underline{1}))$?

- (a) $\text{if}(\text{zero?}(\text{minus}(\text{plus}(\underline{2}, \underline{1}), \underline{3})), \text{times}(\underline{3}, \underline{5}), \text{plus}(\underline{1}, \underline{1}))$
- (b) $\text{if}(\text{zero?}(\underline{0}), \text{times}(\underline{3}, \underline{5}), \text{plus}(\underline{1}, \underline{1}))$
- (c) $\text{if}(\text{minus}(\text{plus}(\underline{2}, \underline{1}), \underline{3}), \text{times}(\underline{3}, \underline{5}), \text{plus}(\underline{1}, \underline{1}))$
- (d) $\text{if}(\text{zero?}(\text{plus}(\underline{2}, \underline{1})), \text{times}(\underline{3}, \underline{5}), \text{minus}(\underline{3}, \underline{3}))$

Exercise 5 (This or That). For each of the following pairs of a context and a sub-expression, identify which ones are valid or invalid decompositions of the expression

$$\text{if}(\text{zero?}(\text{times}(\text{plus}(\underline{3}, \underline{4}), \text{minus}(\underline{1}, \underline{1}))), \text{minus}(\underline{5}, \underline{3}), \text{div}(\underline{6}, \underline{2}))$$

according to the grammar of E in small-step operational semantics. (There may be multiple valid decompositions.)

- (a) $\text{if}(\square, \text{minus}(\underline{5}, \underline{3}), \text{div}(\underline{6}, \underline{2}))$ **and** $\text{zero?}(\text{times}(\text{plus}(\underline{3}, \underline{4}), \text{minus}(\underline{1}, \underline{1})))$
- (b) $\text{if}(\text{zero?}(\text{times}(\text{plus}(\underline{3}, \underline{4}), \text{minus}(\underline{1}, \underline{1}))), \square, \text{div}(\underline{6}, \underline{2}))$ **and** $\text{minus}(\underline{5}, \underline{3})$
- (c) $\text{if}(\text{zero?}(\text{times}(\text{plus}(\underline{3}, \underline{4}), \text{minus}(\underline{1}, \underline{1}))), \text{minus}(\underline{5}, \underline{3}), \square)$ **and** $\text{div}(\underline{6}, \underline{2})$
- (d) $\text{if}(\text{zero?}(\text{times}(\square, \text{minus}(\underline{1}, \underline{1}))), \text{minus}(\underline{5}, \underline{3}), \text{div}(\underline{6}, \underline{2}))$ **and** $\text{plus}(\underline{3}, \underline{4})$
- (e) $\text{if}(\text{zero?}(\text{times}(\text{plus}(\underline{3}, \underline{4}), \square)), \text{minus}(\underline{5}, \underline{3}), \text{div}(\underline{6}, \underline{2}))$ **and** $\text{minus}(\underline{1}, \underline{1})$

Exercise 6 (Show Your Work). Write down the sequence of reduction steps using the rules of the small-step operational semantics for simplifying to its final result.

$$\text{if} \left(\text{zero?} \left(\text{times}(\text{plus}(\underline{1}, \underline{2}), \text{minus}(\underline{3}, \underline{3})), \text{div}(\underline{7}, \underline{3}), \text{minus}(\underline{8}, \underline{2}) \right) \right)$$